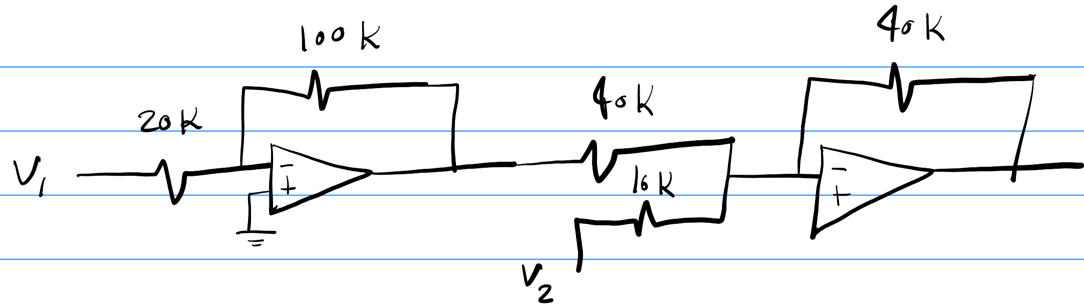
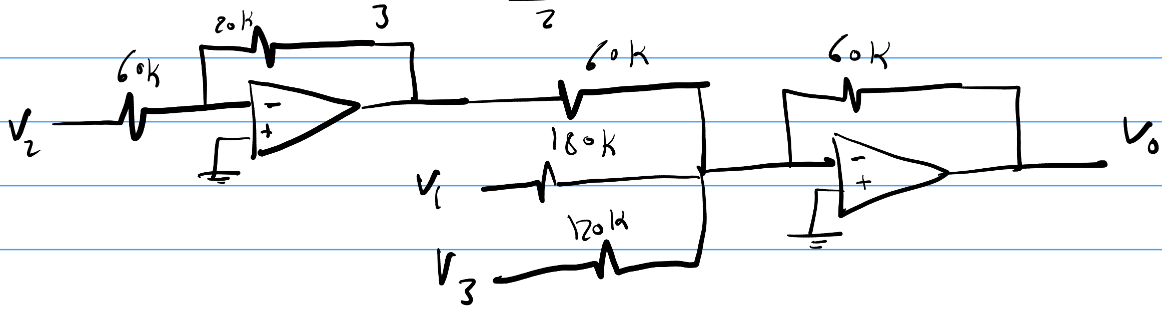
$$V_o = 5V_1 - 2V_2$$

1



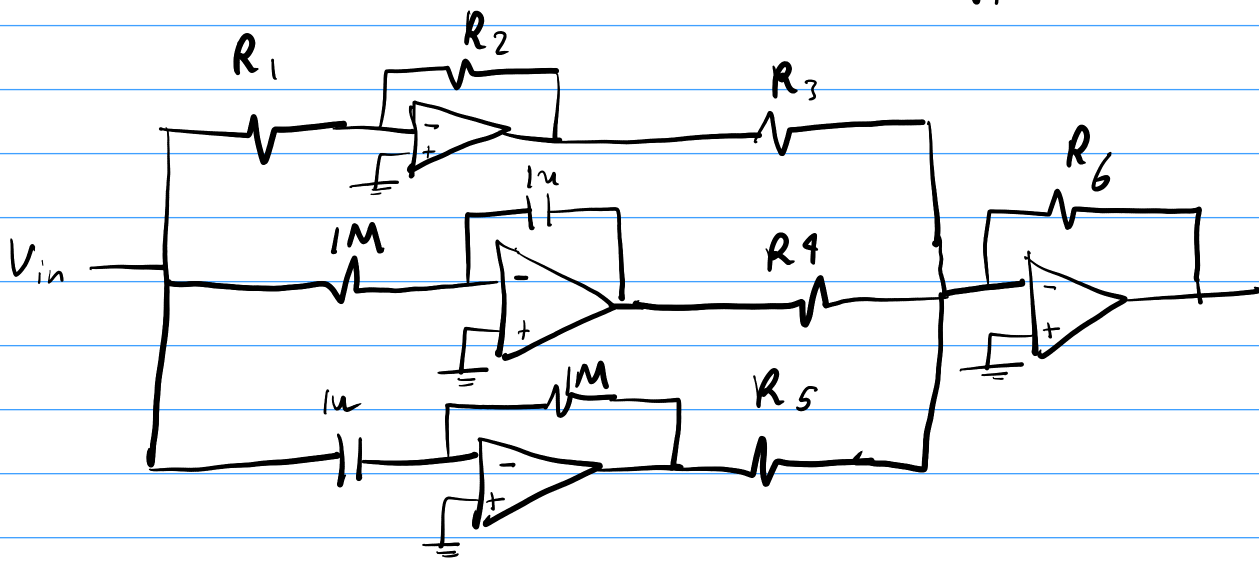
2

$$V_0 = \frac{V_2 - V_1}{3} - \frac{V_3}{2}$$



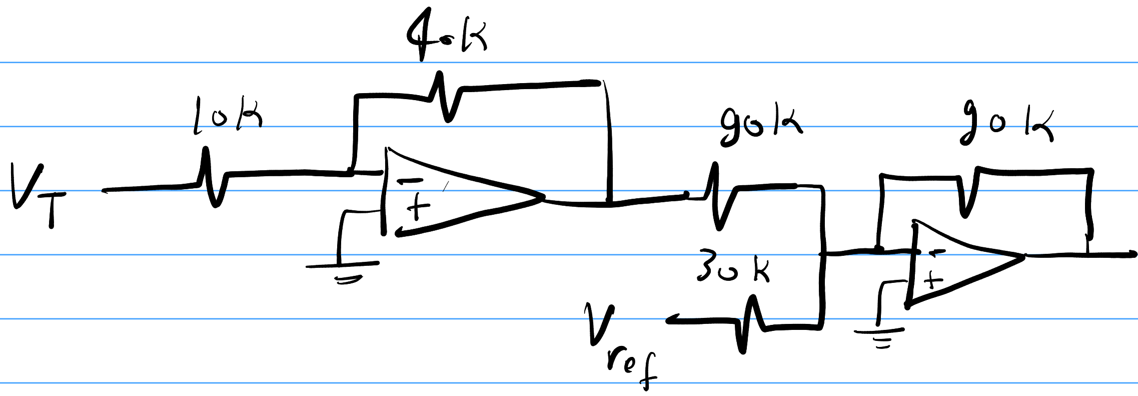
3

$$V_o = K_p V_{in} + K_i \int V_{in}\,dt + K_d \frac{dV_{in}}{dt}$$



$$K_p = \frac{R_6}{R_3} \times \frac{R_2}{R_1}, \quad K_i = -\frac{R_6}{R_4}, \quad K_d = -\frac{R_6}{R_5}$$

$\boxed{4}$



$\boxed{5}$

$$10\ddot{V} + 200\dot{V} + 100V = 250$$

$$10\ddot{V} = 250 - 200\dot{V} - 100V$$